

Appendix B

Multimodal Mathematical function to optimize with GA

Shekel family functions

Definition

Shekel5, Shekel7, Shekel10 functions: $f_{4,m}(x_1, \dots, x_4) = -\sum_{i=1}^m \left(\sum_{j=1}^4 (x_j - a_{ij})^2 + c_i \right)^{-1}$, $m = 5, 7, 10$.

The constants c_i and a_{ij} are given in Table 1.

Properties

- $f_{4,5}(4.00004, 4.00013, 4.00004, 4.00013) = -10.1532$;
- Global minima: $f_{4,7}(4.00057, 4.00069, 3.99949, 3.99961) = -10.4029$;
- $f_{4,10}(4.00075, 4.00059, 3.99966, 3.99951) = -10.5364$;

Search domain

- $D_4 = \{(x_1, \dots, x_4) \in R^4 : -15 \leq x_i \leq 20, i = 1, \dots, 4\}$

Table 1. Data for *Shekel 5,7,10* functions.

i	a_{ij}				c_i
1	4.0	4.0	4.0	4.0	0.1
2	1.0	1.0	1.0	1.0	0.2
3	8.0	8.0	8.0	8.0	0.2
4	6.0	6.0	6.0	6.0	0.4
5	3.0	7.0	3.0	7.0	0.4
6	2.0	9.0	2.0	9.0	0.6
7	5.0	5.0	3.0	3.0	0.3
8	8.0	1.0	8.0	1.0	0.7
9	6.0	2.0	6.0	2.0	0.5
10	7.0	3.6	7.0	3.6	0.5

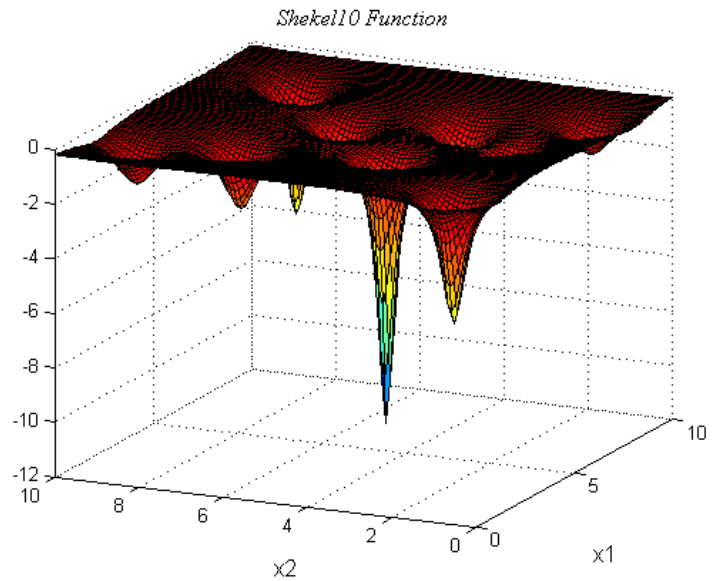


Fig.1. *Shekel10* function.

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